

Figure 1: An encrypted partition with an ext4 file system

```
# cryptsetup luksFormat /dev/vdb1
```

3. Unlock the encrypted volume and assign it a logical volume, as follows:

```
# cryptsetup luksOpen /dev/vdb1 name
```

4. Create a file system in the decrypted volume, using the following command:

```
# mkfs -t ext4 /dev/mapper/name
```

As shown in Figure 1, the partition has been encrypted and opened and, finally, a file system is associated with the partition.

5. Create a mount point for the file system, mount it, and then access its contents as follows:

```
# mkdir /secret
# mount /dev/mapper/name /secret
```

We can verify the mounted partition using the `df -h` command, as shown in Figure 2.

6. When finished, unmount the file system and then lock the encrypted volume, as follows:

```
# umount /secret
```

Note: The directory should be unmounted before closing the LUKS partition. After the partition has been closed, it will be locked. This can be verified using the `df -h` command, as shown in Figure 2.

```
# cryptsetup luksClose name
```

How to persistently mount encrypted partitions

If a LUKS partition is created during installation, normal system operation prompts the user for the LUKS passphrase at boot time. This is fine for a laptop, but not for servers that may need to be able to reboot, unattended.

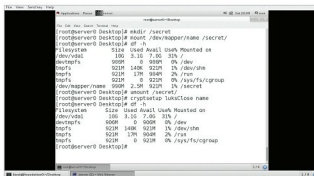


Figure 2: The encrypted partition has been locked and verified

To boot a server with an encrypted volume unattended, a file must be created with a LUKS key that will unlock the encrypted volume. This file must reside on an unencrypted file system on the disk. Of course, this presents a security risk if the file system is on the same disk as the encrypted volume, because theft of the disk would include the key needed to unlock the encrypted volume. Typically, the file with the key is stored on removable media such as a USB drive.

Here are the steps to be taken to configure a system to persistently mount an encrypted volume without human intervention.

1. First, locate or generate a key file. This is typically created with random data on the server and kept on a separate storage device. The key file should take random input from `/dev/urandom`, and generate our output `/root/.key.txt` with a block size of 4096 bytes as a single count of random numbers.

```
# dd if=/dev/urandom of=/root/.key.txt bs=4096 count=1
```

Make sure it is owned by the root user and the mode is 600, as follows:

```
# chmod 600 /root/.key.txt
```

Add the key file for LUKS using the following command:

```
# cryptsetup luksAddKey /dev/vda1 /root/.key.txt
```

Provide the passphrase used to unlock the encrypted volume when prompted.

2. Create an `/etc/crypttab` entry for the volume. `/etc/crypttab`



Figure 3: A key file has been generated and added to the LUKS partition